Project Report: Cloud-Native Banking API Platform - Java Backend Developer

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Family:	Java Backend Developer
Level:	Advanced
Chapters:	6
Source:	CodeQuest project specification (official docs + industry patterns)

Summary

Multi-service platform on Kubernetes with virtual threads, circuit breakers, and distributed tracing.

Chapters

Track: Project-Intro

Chapter 1: Project Overview [Advanced]

Chapter focus: Project Overview *(Level: Advanced)**

Cloud-Native Banking API Platform is a advanced-level end-to-end project for Java Backend Developer. Duration: 6-8 weeks. Multi-service platform on Kubernetes with virtual threads, circuit breakers, and distributed tracing. Tools: Java 21, K8s, Helm, OpenTelemetry.**

Code Reference:

```
Objective: Deploy 3 services on Kubernetes with Helm
Objective: Enable Java 21 virtual threads
Objective: Implement Resilience4j circuit breakers
Objective: End-to-end tracing with OpenTelemetry
```

What it shows: Lists measurable outcomes before you start building.

Topic guide

What it actually is
A project overview defines scope, tools, and success criteria.

When to use it
At project kickoff.

Where to use it
Java Backend Developer portfolios and CodeQuest project submissions.

Real use example
A candidate lists this project on a Java Backend Developer resume with a demo link.

Track: Project-Phase

Chapter 2: Phase: K8s Setup [Advanced]

Chapter focus: Phase: K8s Setup *(Level: Advanced)**

Helm charts and namespaces. Complete these tasks: Dev/staging/prod; Ingress + TLS. Deliverable for this phase: Cluster running.

Code Reference:

```
- [ ] Dev/staging/prod
- [ ] Ingress + TLS
```

What it shows: Checklist format helps track phase completion.

Topic guide

What it actually is
Phase K8s Setup is one step in the full project lifecycle.

When to use it
During week 1 of the project timeline.

Where to use it
Java Backend Developer agile sprints and coursework milestones.

Real use example
Team demo shows the Cluster running to the teacher.

Chapter 3: Phase: Virtual Threads [Advanced]

Chapter focus: Phase: Virtual Threads *(Level: Advanced)**

High-concurrency payment endpoints. Complete these tasks: Structured concurrency; Load test. Deliverable for this phase: JMeter report.

Code Reference:

```
- [ ] Structured concurrency
- [ ] Load test
```

What it shows: Checklist format helps track phase completion.

Topic guide

What it actually is
Phase Virtual Threads is one step in the full project lifecycle.

When to use it
During week 2 of the project timeline.

Where to use it
Java Backend Developer agile sprints and coursework milestones.

Real use example
Team demo shows the JMeter report to the teacher.

Chapter 4: Phase: Resilience [Advanced]

Chapter focus: Phase: Resilience *(Level: Advanced)**

Circuit breakers and retries. Complete these tasks: Resilience4j config; Fallback handlers. Deliverable for this phase: Chaos test video.

Code Reference:

```
- [ ] Resilience4j config
- [ ] Fallback handlers
```

What it shows: Checklist format helps track phase completion.

Topic guide

What it actually is
Phase Resilience is one step in the full project lifecycle.

When to use it
During week 3 of the project timeline.

Where to use it
Java Backend Developer agile sprints and coursework milestones.

Real use example
Team demo shows the Chaos test video to the teacher.

Chapter 5: Phase: Tracing [Advanced]

Chapter focus: Phase: Tracing *(Level: Advanced)**

OpenTelemetry across services. Complete these tasks: Trace IDs in logs; Jaeger UI. Deliverable for this phase: Trace screenshot walkthrough.

Code Reference:

```
- [ ] Trace IDs in logs
- [ ] Jaeger UI
```

What it shows: Checklist format helps track phase completion.

Topic guide

What it actually is
Phase Tracing is one step in the full project lifecycle.

When to use it
During week 4 of the project timeline.

Where to use it
Java Backend Developer agile sprints and coursework milestones.

Real use example
Team demo shows the Trace screenshot walkthrough to the teacher.

Track: Project-Deliverables

Chapter 6: Final Deliverables and Review [Advanced]

Chapter focus: Final Deliverables and Review *(Level: Advanced)**

Submit all final deliverables: Helm charts; Load test report; Jaeger traces; Architecture doc. Skills demonstrated: Kubernetes, Observability, Performance, Resilience. Prepare a 5-minute walkthrough video and README for CodeQuest teacher marking.

Code Reference:

```
Deliverable: Helm charts
Deliverable: Load test report
Deliverable: Jaeger traces
Deliverable: Architecture doc
```

What it shows: Final checklist ensures nothing is missing at submission.

Topic guide

What it actually is
Deliverable review confirms end-to-end project completion.

When to use it
Before deadline and teacher review.

Where to use it
CodeQuest project gallery and interviews.

Real use example
A student submits Helm charts and passes the rubric.